Neoplasia

Molecular Basis of Cancer (LG9)

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Learning outlines:

Carcinogenesis:

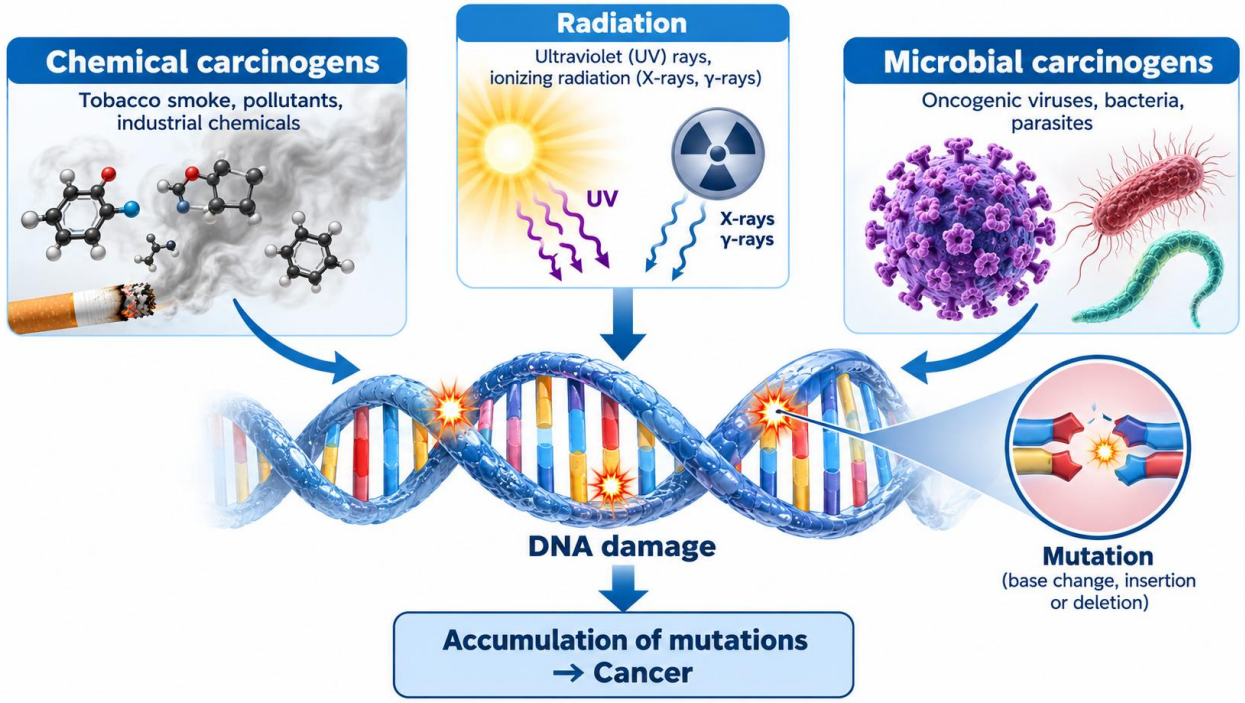
A- Proto-oncogenes

B- Tumour suppressor genes

C- Genes that regulate programmed cell death (apoptosis)

D-DNA repair gene

Carcinogenesis is a multistep process



- Cancer develops by accumulation of non-lethal genetic alterations that can be inherited or acquired.
- Transformed cells proliferate to form a mass.
- A tumor usually represents clonal expansion of one transformed progenitor cell (i.e., tumors are monoclonal).
- Progression reflects accumulation of further mutations.

Carcinogenesis

Normally, there are **four classes of cell cycle regulatory genes** which control cell proliferation.

A. Growth-promoting proto oncogenes.

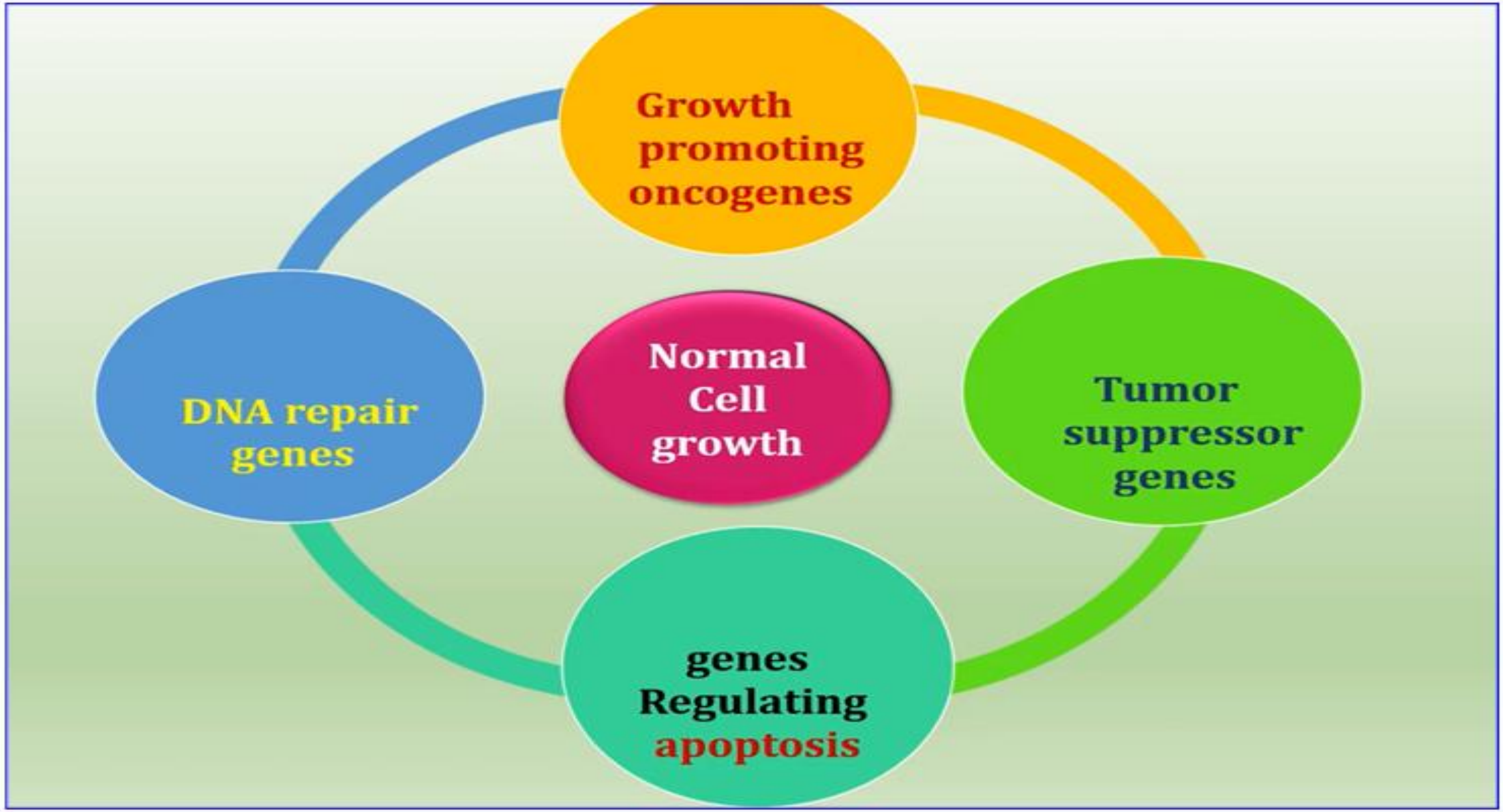
B. Growth-inhibiting tumor suppressor genes.

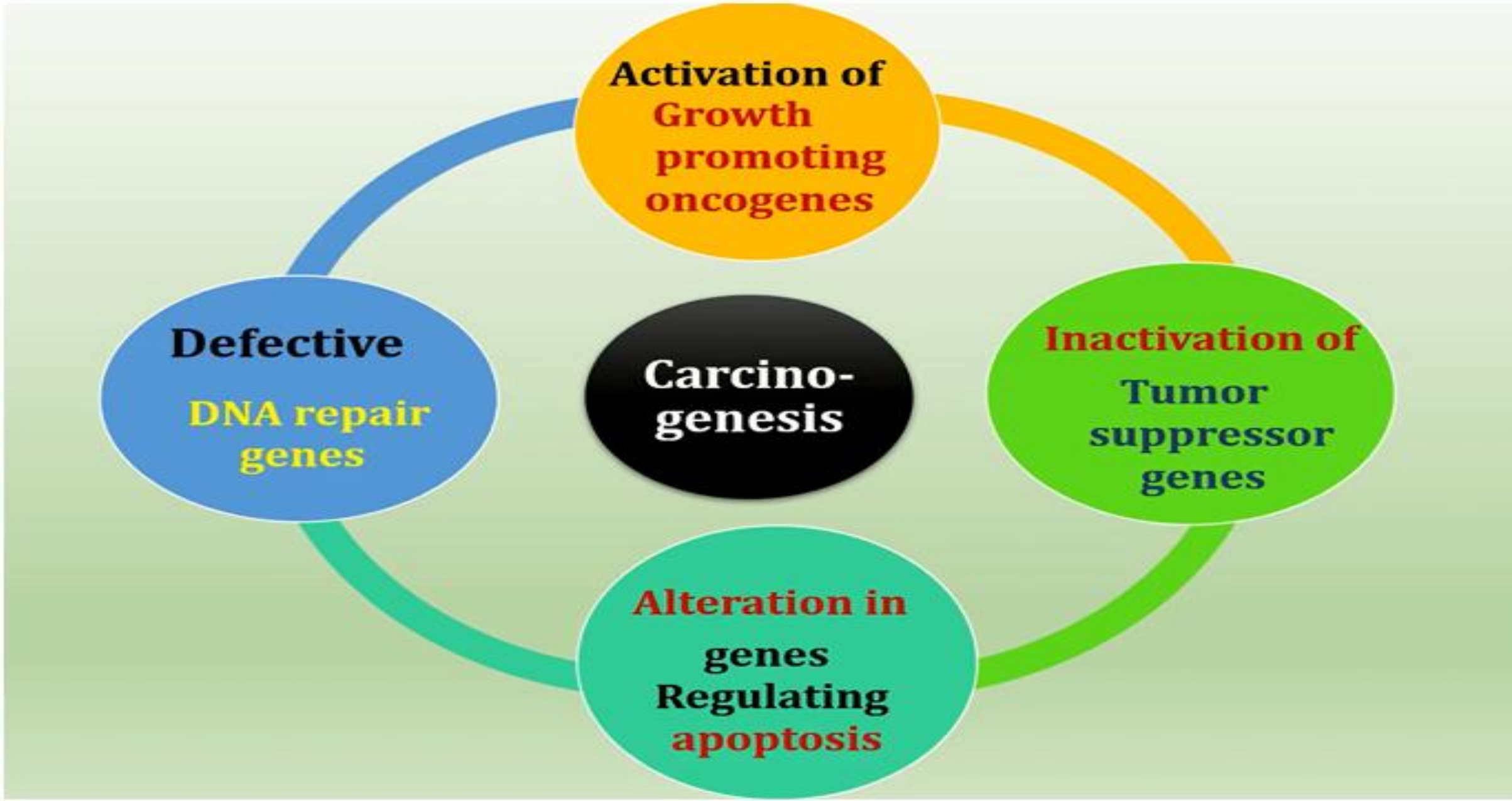
C. Genes that regulate apoptosis.

D. DNA repair genes

Carcinogenesis

- Mutant alleles of **proto-oncogenes** are called **oncogenes**.
- They are considered dominant because mutation of a single allele can lead to cellular transformation.
- Both normal alleles of tumor suppressor genes must be damaged for transformation to occur, referred to as recessive oncogenes.
- Genes that regulate apoptosis may be dominant, as are proto-oncogenes, or they may behave as recessive .

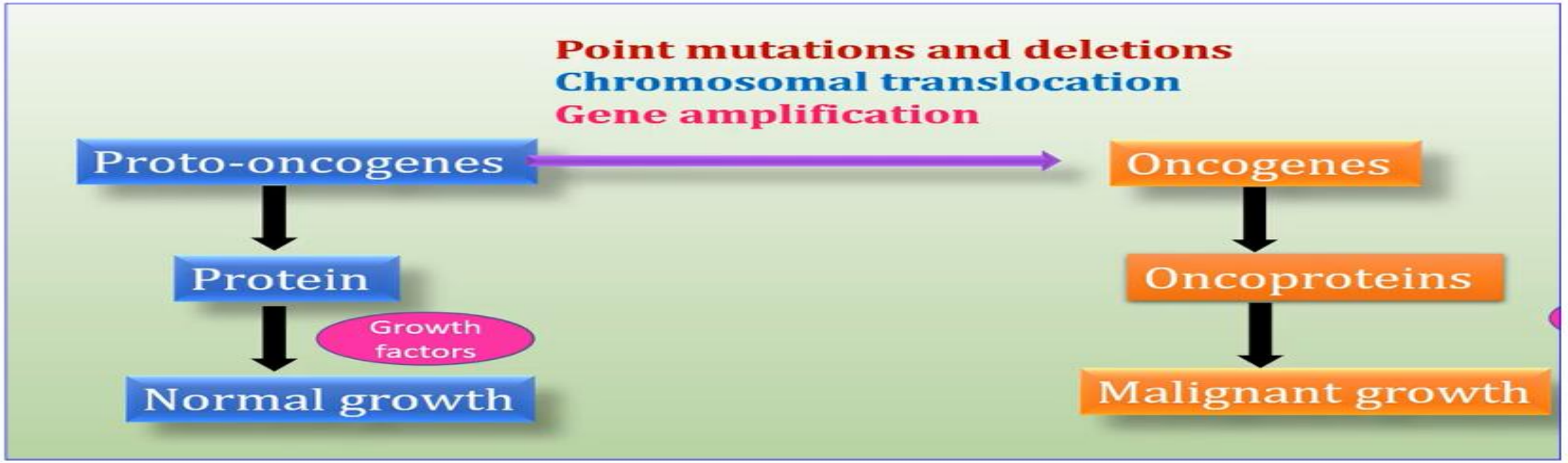

Growth promoting oncogenes
DNA repair genes
Normal Cell growth
Tumor suppressor genes
genes Regulating apoptosis


Activation of
Growth
promoting
oncogenes
Inactivation of
Tumor
suppressor
genes
Alteration in
genes
Regulating
apoptosis
Defective
DNA repair
genes
Carcino-
genesis

A. Proto-oncogenes:

Proto-oncogenes are present normally and they activate cell proliferation in an organised way, when these normal Proto-oncogenes become abnormal (due to mutation), then we call them **Oncogenes.**

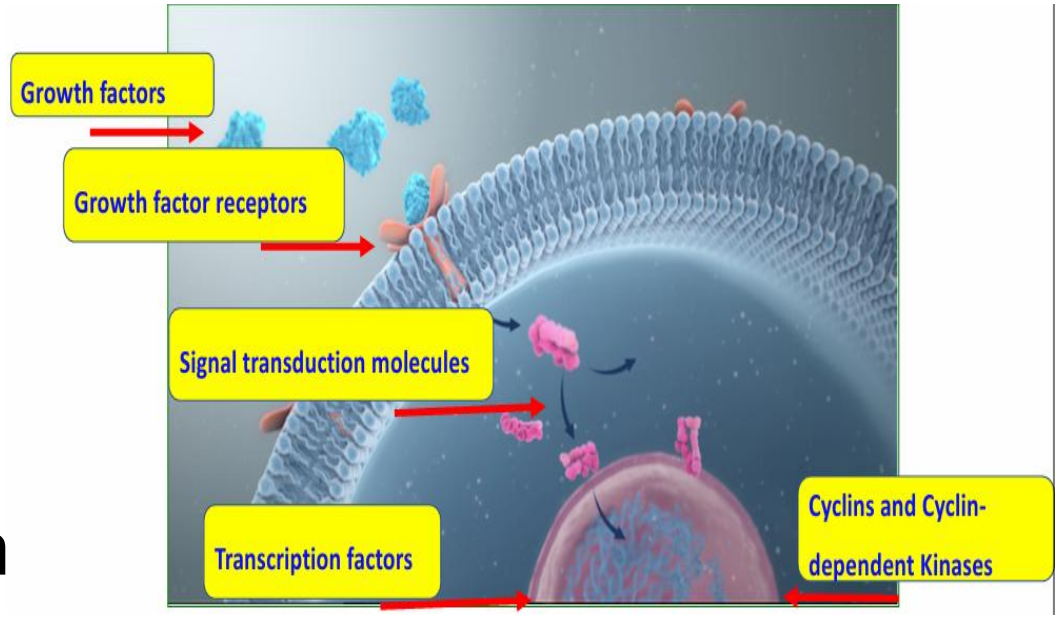
Oncogens promote cell growth in the absence of normal mitogenic signals



A. Proto-oncogenes:

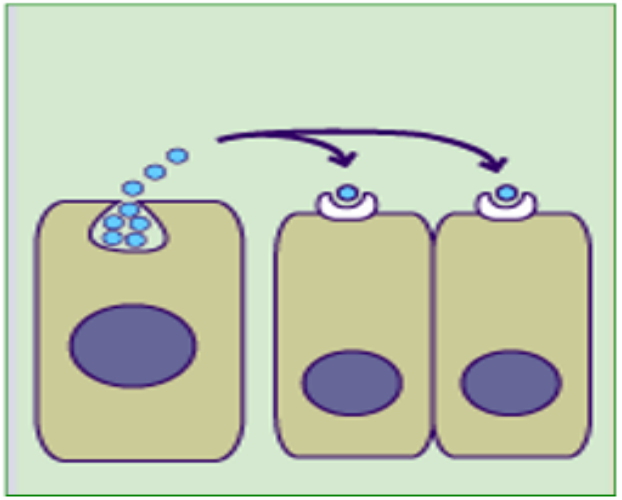
The abnormal genes stimulate cell proliferation through one of **five** mechanisms:

1. Growth factors
2. Growth factor receptors
3. Signal transduction molecules
4. Transcription factors
5. Cyclins and Cyclin-dependent Kina

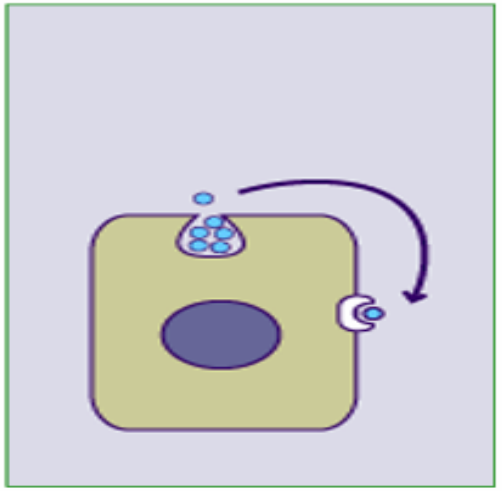


A. Proto-oncogenes:

1. Growth factors: It may increase the growth factors that stimulate cell proliferation.



Normal cells (paracrine signaling).
Signals sent from outside the cell to tell it when it is time to divide.



Tumor cells (autocrine signaling). Cells pump out excessive growth signaling molecules, forcing continuous replication.

A. Proto-oncogenes:

2-Growth Factor Receptors: It may amplify the receptors for growth factors which are present on the surface of cells. Mutated receptors stay permanently switched "on," signaling downstream even when no growth factors are present.

- **RET mutation:** Directly linked to **medullary thyroid carcinoma**.
- **ERBB1:** Associated with **squamous cell carcinomas of the lung**, as well as **head and neck tumors**.
- **ERBB2** (also frequently referred to as ***HER-2/NEU***): Associated with malignancies of the **Breast, Ovary, LUNG, and STOMACH**.
- **c-KIT (CD117) mutation:** Characteristically drives **GIST** (Gastrointestinal Stromal Tumor).

Clinical Applications & Targeted Drug Therapy:
ERBB2 (HER-2/NEU): Targeted using the monoclonal antibody **Trastuzumab (Herceptin)**.
c-KIT: Targeted using the tyrosine kinase inhibitor **Imatinib (Glivec)**.

A. Proto-oncogenes:

3-Signal transduction molecules:

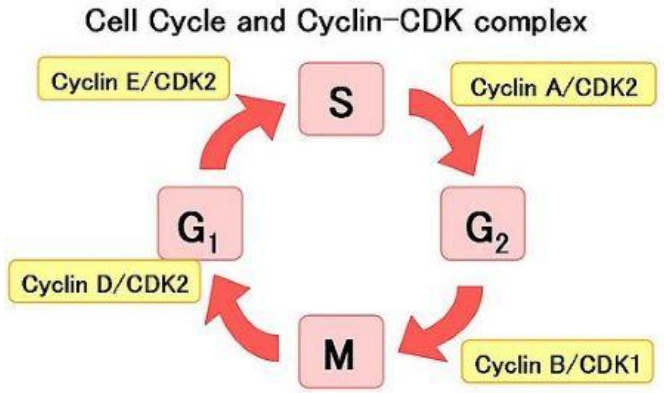
- Messenger proteins (like RAS) that pass the "divide" signal from the receptor to the nucleus.
- **It enhance the signalling pathway inside the cell** which stimulate proliferation (even in the absence of incoming signals).
- There are different types of RAS, the most important is KRAS which is found in many cancers, especially colon cancer.

4-Transcription Factors:

- Nuclear proteins (such as Myc) that activate specific genes required to trigger the cell cycle.
- Excessive production of these factors forces transcription of growth genes continuously (nonstop).
- Examples: the c-MYC proto-oncogene translocation seen in Burkett's lymphoma.

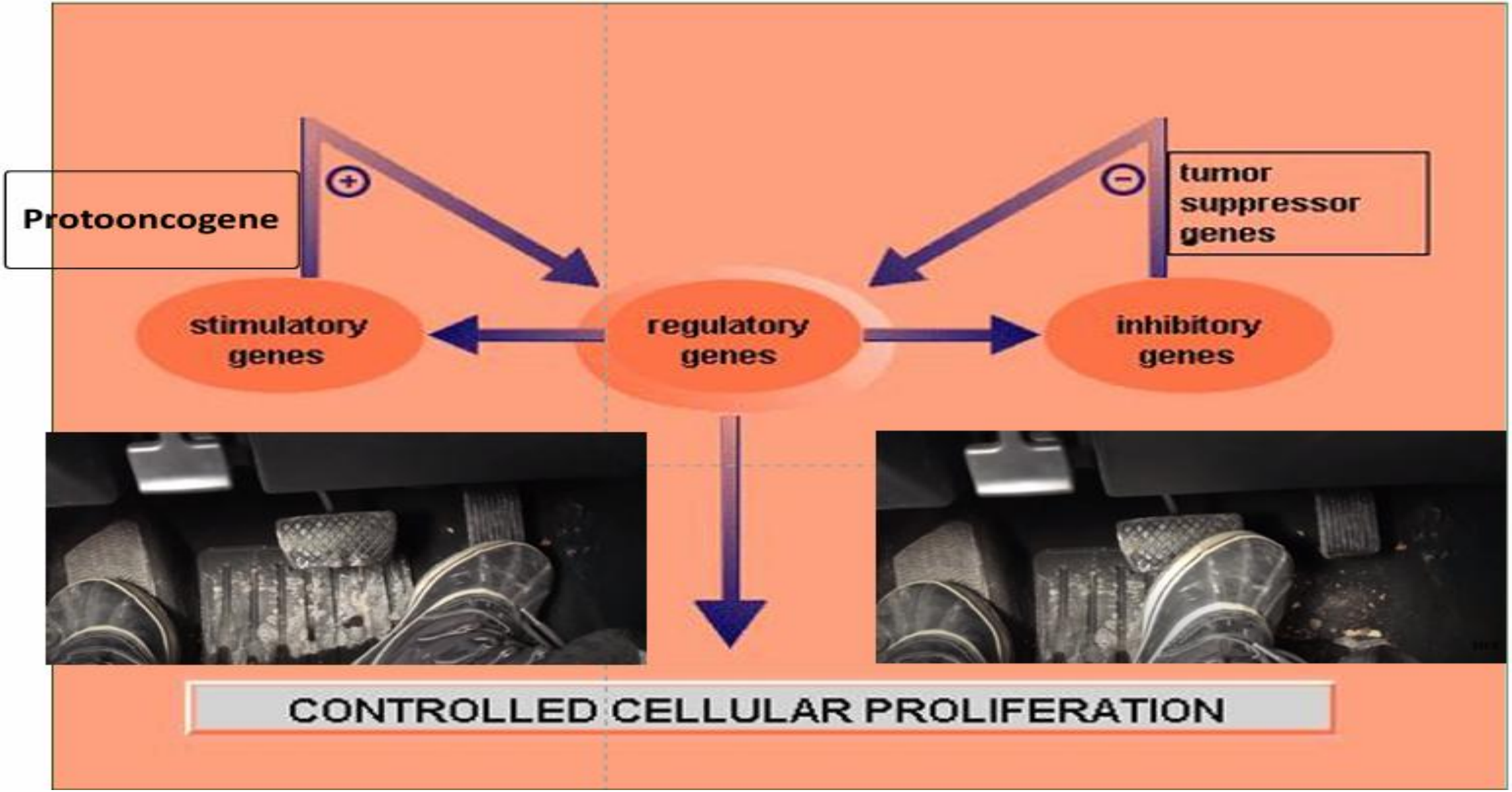
5. Cyclins and Cyclin-dependent Kinases:

- **Cyclins:** Regulatory proteins whose levels rise and fall in rhythmic waves throughout different stages of the cell cycle.
- **Cyclin-dependent Kinases (CDKs):** Enzyme partners that remain inactive until they bind with a specific cyclin.
- In oncogenesis, the hyperactivation or amplification of these specific components (such as **Cyclin D** or **CDK4/6**) allows a tumor cell to autonomously drive its own replication cycle independent of external growth instructions.



B. Tumour suppressor genes:

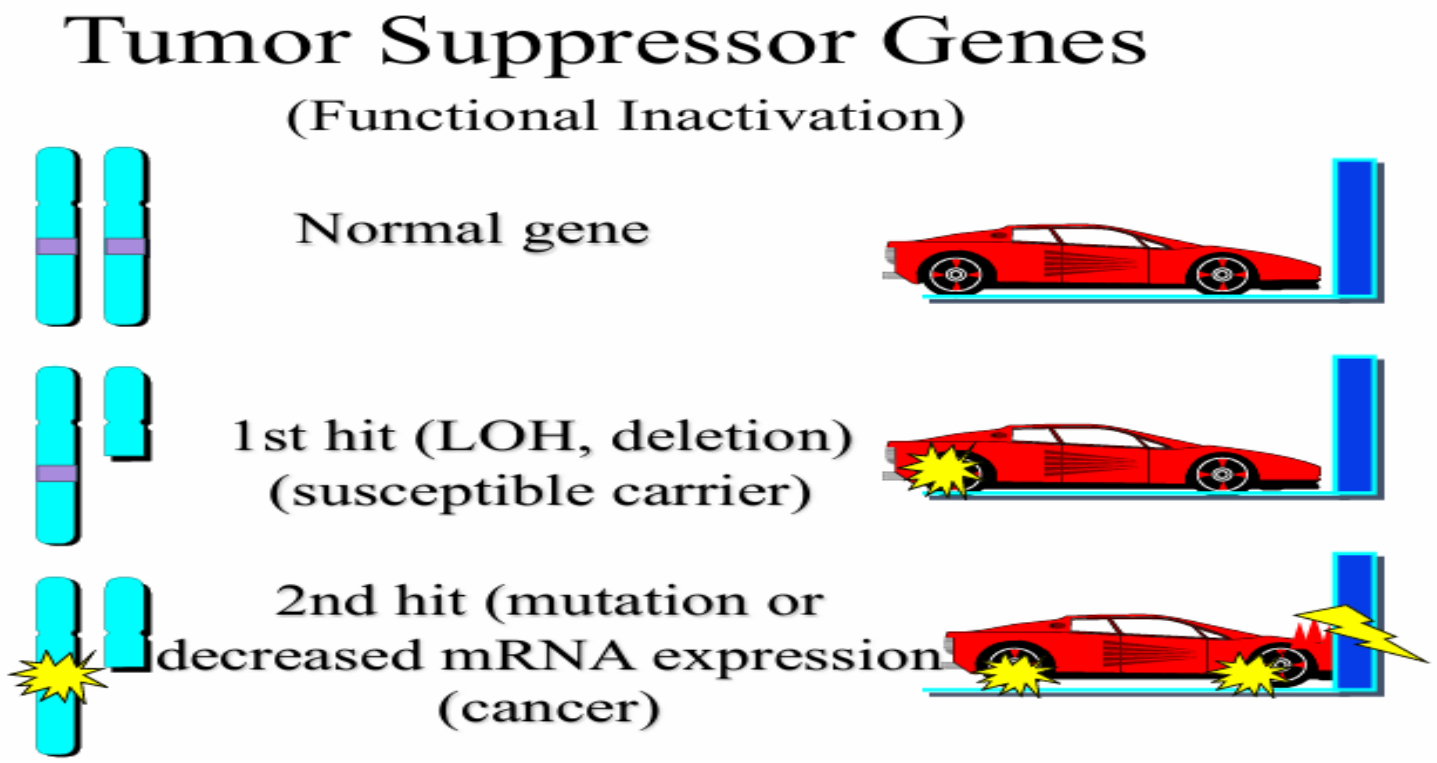
- These genes are present normally and their function is to reduce the rate of cell proliferation.
- They essentially act as the **"brakes" of a cell**.
- When these genes function properly, they prevent cells from growing and multiplying uncontrollably.
- Mutations of these genes lead to the loss of mechanisms that inhibit unwanted cell proliferation.
- Example are the RB gene (Retinoblastoma gene)and p53.


Protooncogene
tumor suppressor genes
stimulatory genes
regulatory genes
inhibitory genes
CONTROLLED CELLULAR PROLIFERATION

B. Tumour suppressor genes:

The "Two-Hit Hypothesis"

For a tumor suppressor gene mutation to cause cancer, **both copies (alleles) of the gene must usually be damaged or inactivated**.



RB and p53: key tumor suppressors

TP53 gene

- "Guardian of the genome."
- Senses DNA damage.
- Induces cell-cycle arrest, DNA repair or apoptosis.
- Commonly altered in about 50% of human cancers: Bladder, breast, colorectal, and lung cancers; Li-Fraumeni syndrome..

RB gene

- Prototype tumor suppressor gene.
- Controls G1/S cell-cycle checkpoint.
- Retinoblastoma requires loss of both normal alleles (two-hit model).

C- Genes that regulate programmed cell death (apoptosis)

- Mutations of these genes lead to the survival of cells that are destined to die (evasion or escaping of apoptosis).
- The best example is mutation of BCL-2 gene seen in Follicular lymphoma (a type of lymphoma), this mutation make cells capable of evading apoptosis, this lead to gradual accumulation of abnormal cells causing cancer.

D- DNA repair genes:

- These are genes that monitor and supervise the replication of DNA during the cell-cycle.
- If the new copied DNA is abnormal, then the DNA repair genes try to correct the abnormal DNA.
- If the abnormality is not corrected then p53 act and initiate apoptosis.
- Examples of these genes are HNPCC (hereditary non-polyposis colorectal cancer syndrome) and BRCA-1 & BRCA-2 genes causing familial breast and ovarian cancers.

Summary

- Cancer is driven by accumulated genetic and epigenetic alterations.
- Oncogenes act as “accelerators”; tumor suppressors act as “brakes.”
- RB follows the two-hit model; p53 links DNA damage to arrest, repair or apoptosis.
- Defective apoptosis and DNA repair increase survival of abnormal clones.
- Molecular pathology guides diagnosis, prognosis and targeted treatment.

Thank you